Definition

Let $f = \langle \mathsf{st}, \mathbf{A}, \mathbf{B}, \mathbf{C} \rangle$ be a proper LTI system from $l \in \mathbb{N}$ to $m \in \mathbb{N}$. Given factorizations $l = i + k$ and $m = j + k$, we define the proper LTI system $\mathrm{Tr}^{k}_{i,j}\, f$ as

$$\langle \mathsf{st}, \mathbf{A} + \mathbf{B}^{[:,i+1:k]}\mathbf{C}^{[j+1:j+k,:]}, \mathbf{B}^{[:,1:i]}, \mathbf{C}^{[1:j,:]} \rangle$$

where the notation $\mathbf{A}^{[a:b,c:d]}$ means considering rows a to b and columns c to d of matrix $\mathbf{A}$. In case only $:$ is given, all rows/column are considered.